

# Lecture 2

## Production

Macroeconomics (Quantitative)

Econ 101B

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# A Model of Production

- Economists use mathematical **models** to describe and understand the world
- In this lecture, we take our first steps in this direction

# Simple Canonical Model

- One market: Supply and demand:

$$\begin{aligned}q_t &= ap_t + w_t \\q_t &= -bp_t + p_t^{xyz}\end{aligned}$$

- Two equations look very similar
- Which one is the supply curve?
  - If  $a$  and  $b$  are positive, first equation is supply curve because supply curves slope upward

# Components of a Model

1. Endogenous variables
2. Exogenous variables and parameters
3. Equations that relate these variable to each other

# Components of a Model

$$\begin{aligned}q_t &= ap_t + w_t \\q_t &= -bp_t + p_t^{xyz}\end{aligned}$$

- Endogenous variables:  $p_t$  and  $q_t$
- Exogenous variables:  $w_t$  and  $p_t^{xyz}$
- Parameters:  $a$  and  $b$
- If you don't know what the endogenous variables are, you don't know what you are solving for.

# General Equilibrium Model

- In macro, we usually study **general equilibrium** models
- A model is a general equilibrium model if price **and** quantity in **all** markets are endogenous variables
- Today we will study our first general equilibrium model with three markets:
  - Labor market, capital market, goods market

# Roadmap

- Simple general equilibrium models will consist of:
  - Demand and supply curves for each market
- Important caveat #1: Only relative prices matter
  - This implies that we can set one of the prices equal to 1
  - We call this the numeraire
- Important caveat #2: Walras' Law
  - If all but one market is in equilibrium, then the last one is necessarily also in equilibrium.
  - This implies that we only need to write down equations for all but one of the markets

# Roadmap

- To solve a model, you need as many equations as there are endogenous variables
- Example: 3 markets (labor, capital, goods)
  - How many endogenous variables?
  - Five endogenous variables:  $L, w, K, r, Y$ .
    - Price of goods is the numeraire
  - We will need 5 equations
    - E.g.: demand and supply in the labor and capital markets and one more equation to give us output.

# Production Function

- We represent production process by a production function:

$$Y = F(K, L)$$

- Example: How much ice cream can we make if we have  $L$  workers and  $K$  ice cream machines?
- $Y$ : amount of ice cream produced
- $L$ : Number of workers (or hours of work)
- $K$ : Number of machines (amount of capital)
- Inputs or **factors of production**: Labor, Capital

# Properties of Production Functions

- What properties should a reasonable production function satisfy?

$$Y = F(K, L)$$

- Suppose we increase  $L$ 
  - Output increases
  - But at a declining rate
- Suppose we increase  $K$ 
  - Output increases
  - But at a declining rate

# Properties of Production Functions

- What properties should a reasonable production function satisfy?
- Positive marginal products of factors:

$$\frac{\partial F}{\partial L} \geq 0 \text{ and } \frac{\partial F}{\partial K} \geq 0$$

- Diminishing marginal products:

$$\frac{\partial^2 F}{\partial L^2} \leq 0 \text{ and } \frac{\partial^2 F}{\partial K^2} \leq 0$$

(What about  $\partial^2 F / \partial L \partial K$ ?)

# Cobb-Douglas Production Function

- Most commonly used production function

$$Y = AK^a L^{1-a}$$

$$\frac{\partial Y}{\partial L} = (1 - a)AK^a L^{-a} \geq 0 \quad \text{and} \quad \frac{\partial Y}{\partial K} = aAK^{a-1} L^{1-a} \geq 0$$

$$\frac{\partial^2 Y}{\partial L^2} = -a(1 - a)AK^a L^{-a-1} \leq 0 \quad \text{and} \quad \frac{\partial^2 Y}{\partial K^2} = -a(1 - a)AK^{a-2} L^{1-a} \leq 0$$

$$\frac{\partial^2 Y}{\partial L \partial K} = a(1 - a)AK^{a-1} L^{-a} \geq 0$$

# Returns to Scale

- Suppose:

$$Y = AK^a L^{1-a}$$

- What happens if we double all inputs to production?  
(i.e., both  $K$  and  $L$ )

$$\begin{aligned} F(2K, 2L) &= A(2K)^a (2L)^{1-a} \\ &= A2^a K^a 2^{1-a} L^{1-a} \\ &= 2AK^a L^{1-a} \\ &= 2F(K, L) \end{aligned}$$

- Twice as much of every input yields twice as much output
- Production function is **constant returns to scale**

# Returns to Scale

- What is it about  $Y = AK^aL^{1-a}$  that implies constant returns to scale?
  - Exponents on  $K$  and  $L$  sum to one
- If exponents sum to more than one:
  - Increasing returns to scale
- If exponents sum to less than one:
  - Decreasing returns to scale
- Economically, why should we model production as having constant returns to scale at the aggregate level?

# Replication Argument

- Replication argument:
  - If you build two identical factories. You should get twice as much output as from one.
- Counter-arguments?
  - Micro level scale economies that are not fully exhausted
  - Fixed factors: Some inputs cannot be doubled. E.g., land.
    - Not strictly speaking about the production function itself.
    - But has similar implications (diminishing returns to remaining factors)
  - Externalities: There may be either positive or negative spillovers between factories

# Behavior of Firms

- How should we model the behavior of firms?
- Usual assumptions in economics:
  - Individuals maximize utility
  - Firms maximize profits
- Tricky for firms. ... What is a firm?
  - Shareholders put up money and “own” the firm
  - Managers hired by shareholders
  - Employees hired by managers
  - Money borrowed from creditors
- Conflicts of interest left and right!!

# Firm's Problem

- Choose how many workers to hire and machines to rent to maximize profits

$$\max_{K,L} F(K, L) - rK - wL$$

- Profits are production minus cost of inputs
- We set the price of output to 1 (the numeraire)
- $r$ : rental price of machines,  $w$ : wage of workers
- Firm takes  $r$  and  $w$  as given. Why?
  - Competitive markets for labor and capital  
(Simplifying assumption – left wing critique)

# Firm's Problem: Solution

- How do we go about solving the firm's problem?

$$\max_{K,L} F(K, L) - rK - wL$$

1. Choose optimal level of capital
  - Maximize profits with respect to  $K$  treating  $L$  as fixed
2. Choose optimal level of labor
  - Maximize profits with respect to  $L$  treating  $K$  as fixed

# Firm's Problem: Solution

- We are making use of a simple but powerful mathematical result:

$$\max_{x,y} f(x, y)$$

is given by the solution to the following two equations

$$\frac{\partial f(x, y)}{\partial x} = 0$$

$$\frac{\partial f(x, y)}{\partial y} = 0$$

# Firm's Problem: Solution

$$\max_{K,L} F(K, L) - rK - wL$$

- We start by plugging in for  $F(K, L)$ :

$$\max_{K,L} AK^a L^{1-a} - rK - wL$$

# Optimal Choice of Capital by Firm

- Differentiate profit function with respect to  $K$   
(holding  $L$  constant)

$$\frac{\partial}{\partial K} AK^a L^{1-a} - rK - wL = aAK^{a-1} L^{1-a} - r$$

- Set this equal to zero to maximize:

$$aAK^{a-1} L^{1-a} - r = 0$$

$$aAK^{a-1} L^{1-a} = r$$

# Optimal Choice of Labor by Firm

- Differentiate profit function with respect to  $L$   
(holding  $K$  constant)

$$\frac{\partial}{\partial L} AK^a L^{1-a} - rK - wL = (1 - a)AK^a L^{-a} - w$$

- Set this equal to zero to maximize:

$$aAK^a L^{-a} - w = 0$$

$$(1 - a)AK^a L^{-a} = w$$

# Firm's Problem: Solution

- Firm maximizes profits if:

$$aAK^{a-1}L^{1-a} = r$$

$$(1 - a)AK^aL^{-a} = w$$

- Seems a bit inscrutable?
- How can we interpret these conditions?

# Firm's Optimal Choice of Capital

$$aAK^{a-1}L^{1-a} = r$$

- RHS: Price of capital
- LHS:

$$\frac{\partial}{\partial K} Y = \frac{\partial}{\partial K} AK^a L^{1-a} = aAK^{a-1}L^{1-a}$$

- LHS: Marginal product of capital
- Optimal for firms to rent capital to the point where the marginal product of capital is equal to price of capital

# Firm's Optimal Choice of Labor

$$(1 - a)AK^aL^{-a} = w$$

- RHS: price of labor (wage)
- LHS

$$\frac{\partial}{\partial L} Y = \frac{\partial}{\partial L} AK^aL^{1-a} = (1 - a)AK^aL^{-a}$$

- LHS: Marginal product of labor
- Optimal for firms to hire labor to the point where the marginal product of labor is equal to price of labor (the wage)

# Firm's Problem

- Maximize profits:

$$\max_{K,L} AK^a L^{1-a} - rK - wL$$

- Optimal choice of capital implies:

$$aAK^{a-1}L^{1-a} = r$$

- Optimal choice of labor implies:

$$(1 - a)AK^a L^{-a} = w$$

# Math vs. Economics

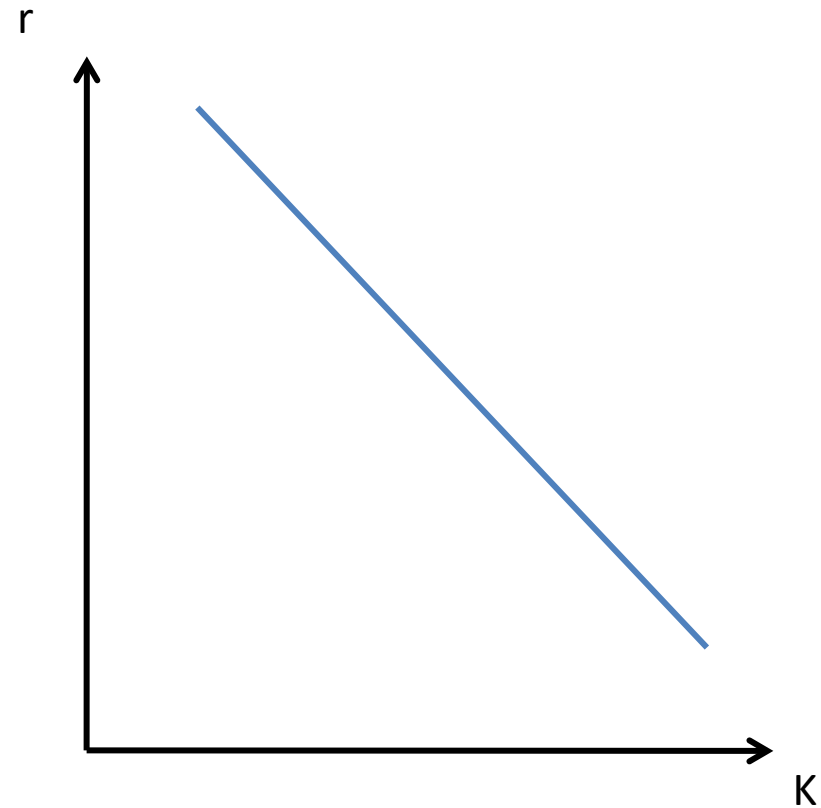
- We can derive optimal behavior by firms in two ways:
  - Mathematically
    - Differentiate profits with respect to capital holding labor fixed
    - Differentiate profits with respect to labor holding capital fixed
  - By economic reasoning:
    - Firm sets MPK equal to price of capital
    - Firm sets MPL equal to price of labor

# Firm Behavior

- Optimal choice of capital yields:

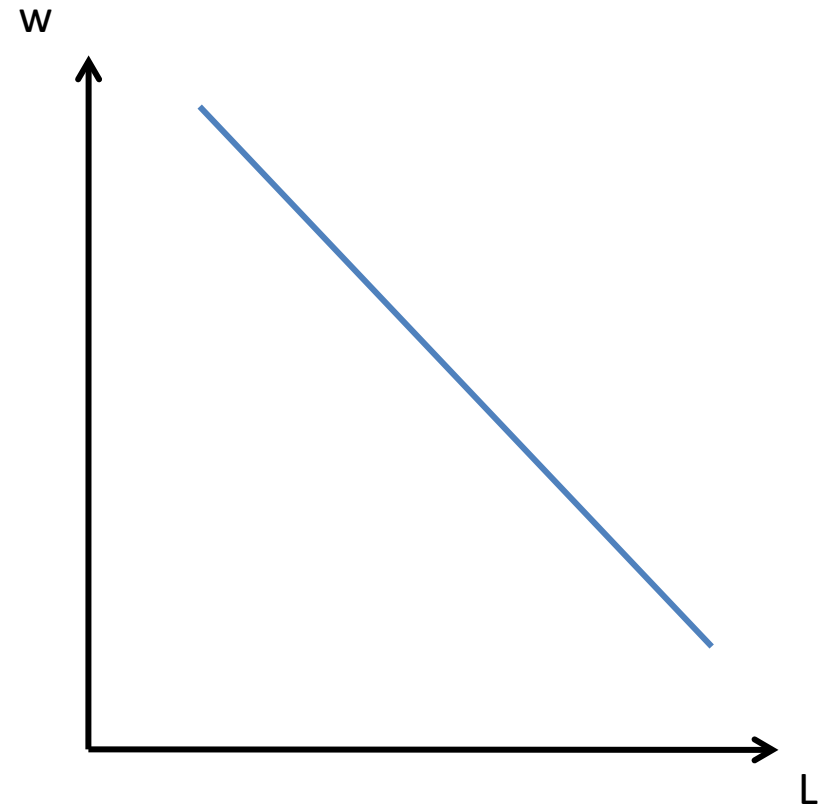
$$aAK^{a-1}L^{1-a} = r$$

- This equation can be represented graphically by a “curve” in (K,r) space
- The curve is all the points in (K,r) space that satisfy the equation (for some set values of other variables, i.e.,  $L, A, a$ )
- Why is this curve downward sloping?
- Does this “curve” have a name?
  - Demand curve in capital market



# Firm Behavior

- Optimal choice of labor yields:  
$$(1 - a)AK^aL^{-a} = w$$
- This equation can also be represented graphically by a “curve” in (L,w) space
- Curve is all the points in (L,w) space that satisfy the equation (for some set values of other variables)
- Does this “curve” have a name?
  - Demand curve in labor market



# Power of Competition

- We assume labor market is perfectly competitive
- This implies that workers are paid their marginal product
  - Workers paid value of what they produce at the margin  
(value of what last worker produces)
- Why aren't they paid less?  
(Why don't firms exploit the workers?)
  - Each firm is held in check by competition with other firms in the labor market (if one pays less, another firm will find it profitable to hire away that firm's workers)

# Power of Competition

- But perhaps perfect competition is not a realistic assumption
- Perhaps we only make this assumption because it simplifies the math
- And then we forget that we make it
- And we start to think markets are efficient in the real world
- And we might even forget about the very possibility that markets might not be competitive
- Daniel Kahneman called this “theory-induced blindness”

# Completing the Model

- We need demand and supply curves for the capital and labor markets + one more equation
- We have labor and capital demand
- What about labor and capital supply?
- Simplifying assumption: labor and capital supplied inelastically

$$K = \bar{K}$$

$$L = \bar{L}$$

# Completing the Model

- Last equation: Something that gives output as a function of other variables
- We can use production function:

$$Y = AK^a L^{1-a}$$

# Model

- Five equations:

$$aAK^{a-1}L^{1-a} = r$$

$$(1 - a)AK^aL^{-a} = w$$

$$K = \bar{K}$$

$$L = \bar{L}$$

$$Y = AK^aL^{1-a}$$

- Five endogenous variables:  $K, L, r, w, Y$

# An Equilibrium

- In economics, the solution to a model is called an **equilibrium**
  - What happens when things play out in the markets under consideration (“when markets clear”)
- Model: System of five equations with five unknowns
- “Solving” the model:
  - Solving for endogenous variables as functions **only** of exogenous variables and parameters
  - Rewriting the system with the endogenous variables on one side of the equations and **only** exogenous variables and parameters on the other side.

# Equilibrium

- “Trivial” solution in our case:

$$K = \bar{K}$$

$$L = \bar{L}$$

$$r = aA\bar{K}^{a-1}\bar{L}^{1-a}$$

$$w = (1 - a)A\bar{K}^a\bar{L}^{-a}$$

$$Y = A\bar{K}^a\bar{L}^{1-a}$$

# How Can We Use The Model?

- The model can help us think about how technology affects workers
- Model assumes full employment (markets clear)
  - Technology will not lead to mass unemployment in the model
- But how does technology affect
  - 1) Wages
  - 2) Labor share of income

# How Does Technology Affect Wages?

- Wages in our model:

$$w = (1 - a)AK^aL^{-a}$$

- An improvement in technology is an increase in  $A$
- How does this affect wages:

$$\frac{\partial w}{\partial A} = (1 - a)K^aL^{-a} > 0$$

- Better technology increases wages

# Labor Share in Our Model

- Labor compensation is  $wL$ . Labor share is  $wL/Y$ .
- From labor demand curve we have

$$w = (1 - a)AK^aL^{-a} = (1 - a)\frac{Y}{L}$$

- Multiplying through by  $L$  and dividing through by  $Y$

$$\frac{wL}{Y} = (1 - a)$$

- Labor share is  $(1 - a)$

# Technology and Workers

- In our model, improved technology:
  - Raises wages
  - Does not affect the labor share of income
- Workers share (proportionately) in benefits of technical change
- Our model is unable to make contact with common anxiety about effects of technology on workers
- Model might (of course) be wrong

# Why a Cobb-Douglas Production Function?

- Cobb-Douglas is tractable
- Arguably more important that Cobb-Douglas implies constant labor share
- Labor share has in fact been remarkably close to constant since WWII

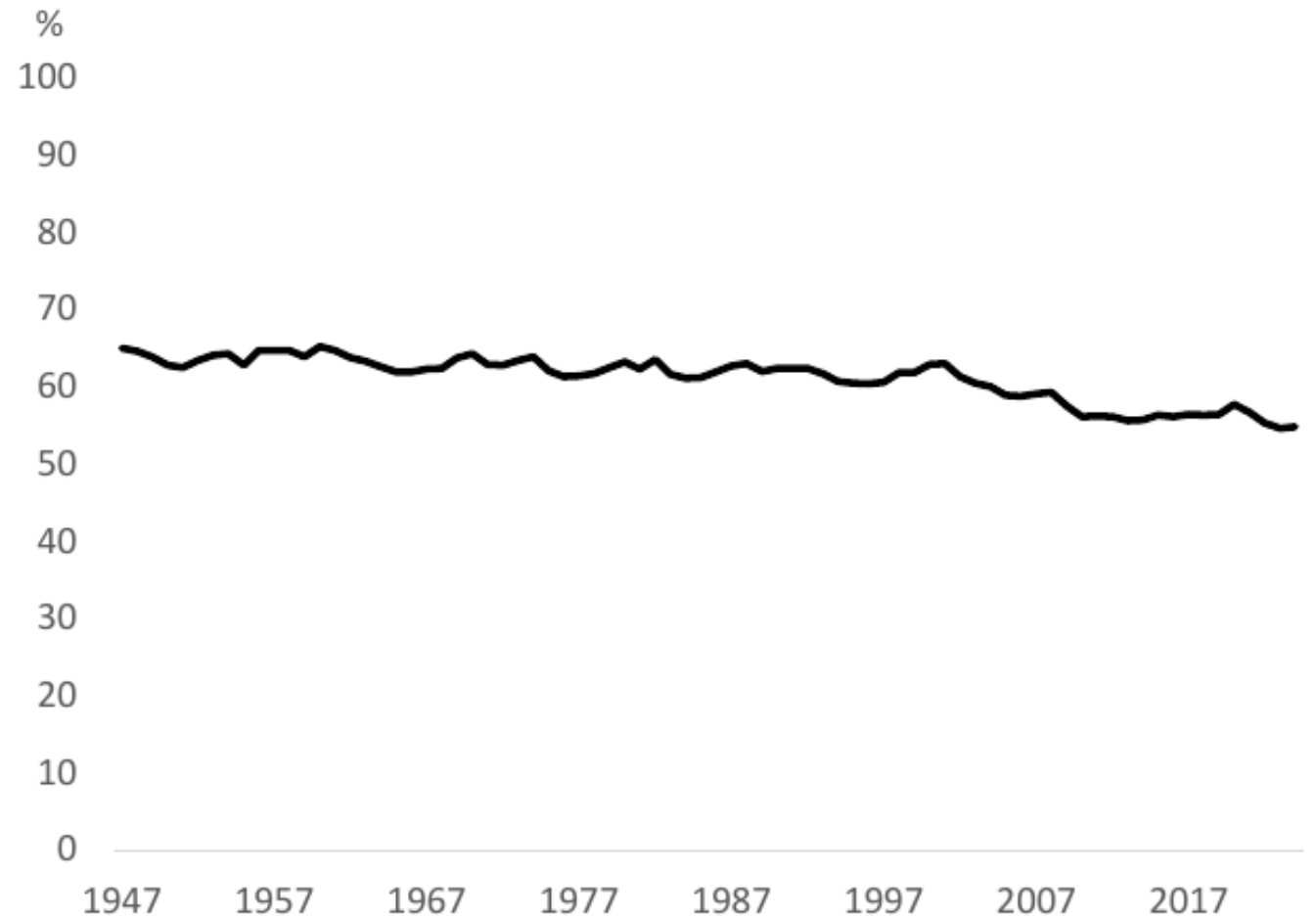


Figure 2: Labor Share in U.S. Nonfarm Business Sector

# Worries about Labor-Saving Technology

- Worries ever since Industrial Revolution that machines will take our jobs
- Labor share constant despite huge amounts of labor-saving technology
- Stark contrast with horses
- But is our luck about to run out?

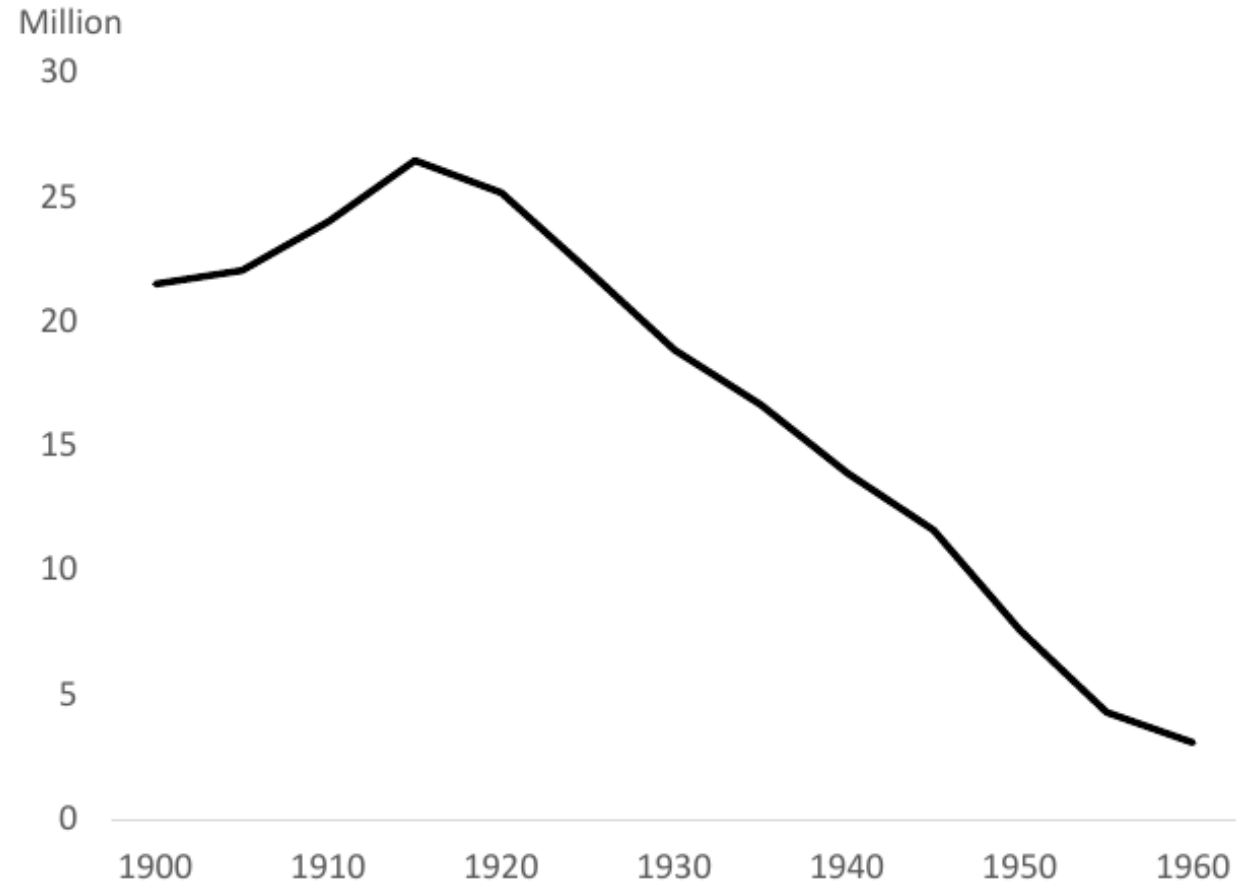


Figure 3: Number of Horses and Mules in the United States, 1900-1960

# Is the Labor Share Falling?

- Closer looks reveals downward trend in labor share
- Are machines finally doing us in?

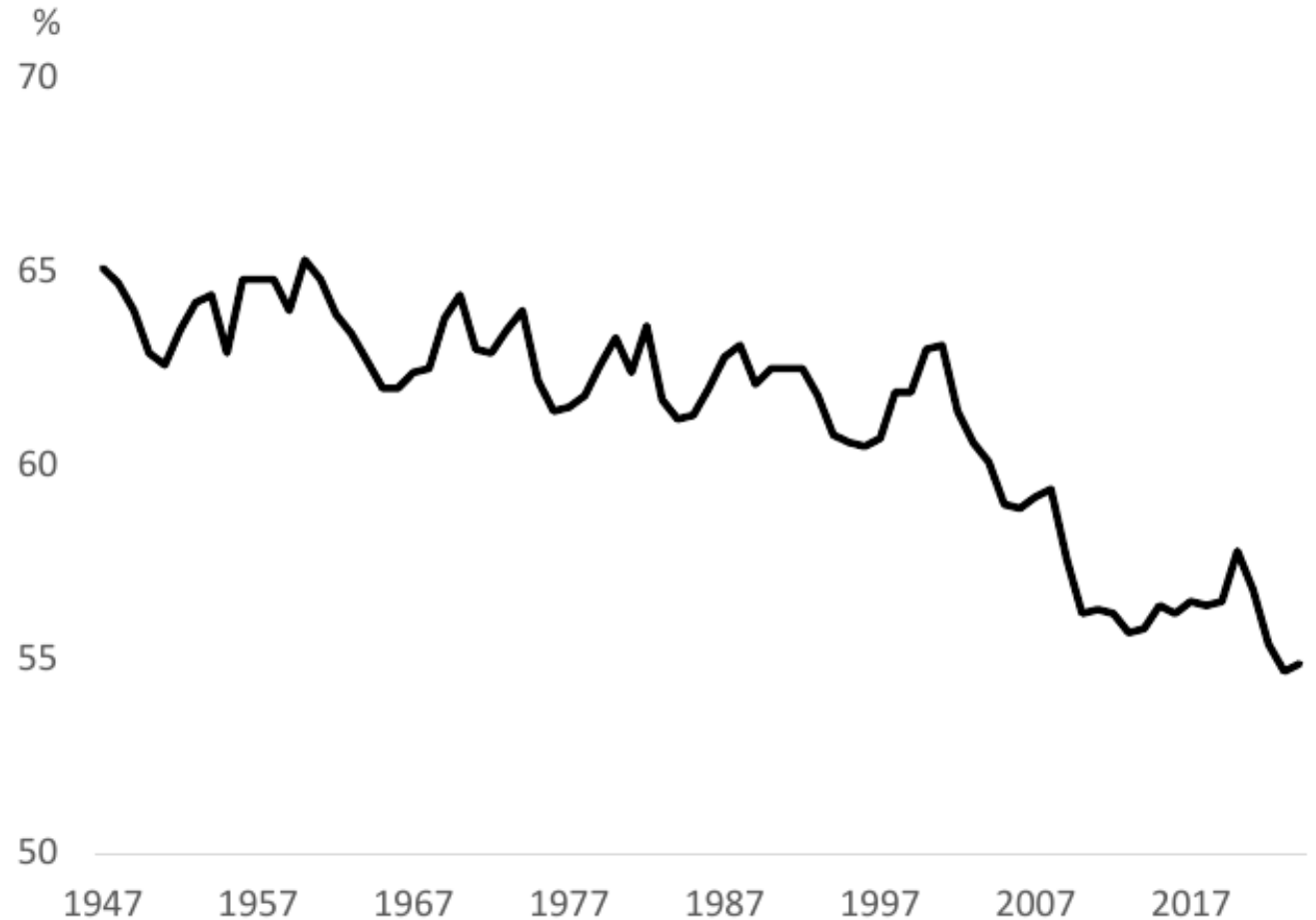


Figure 4: Labor Share in U.S. Nonfarm Business Sector, A Closer Look

# Is the Labor Share Falling?

- Recent research has argued that apparent fall in labor share is partly due to mismeasurement
- Labor share using pre-1999 methods has not fallen
- Difference: Reclassification of intellectual property as capital
- But aren't the new methods better?

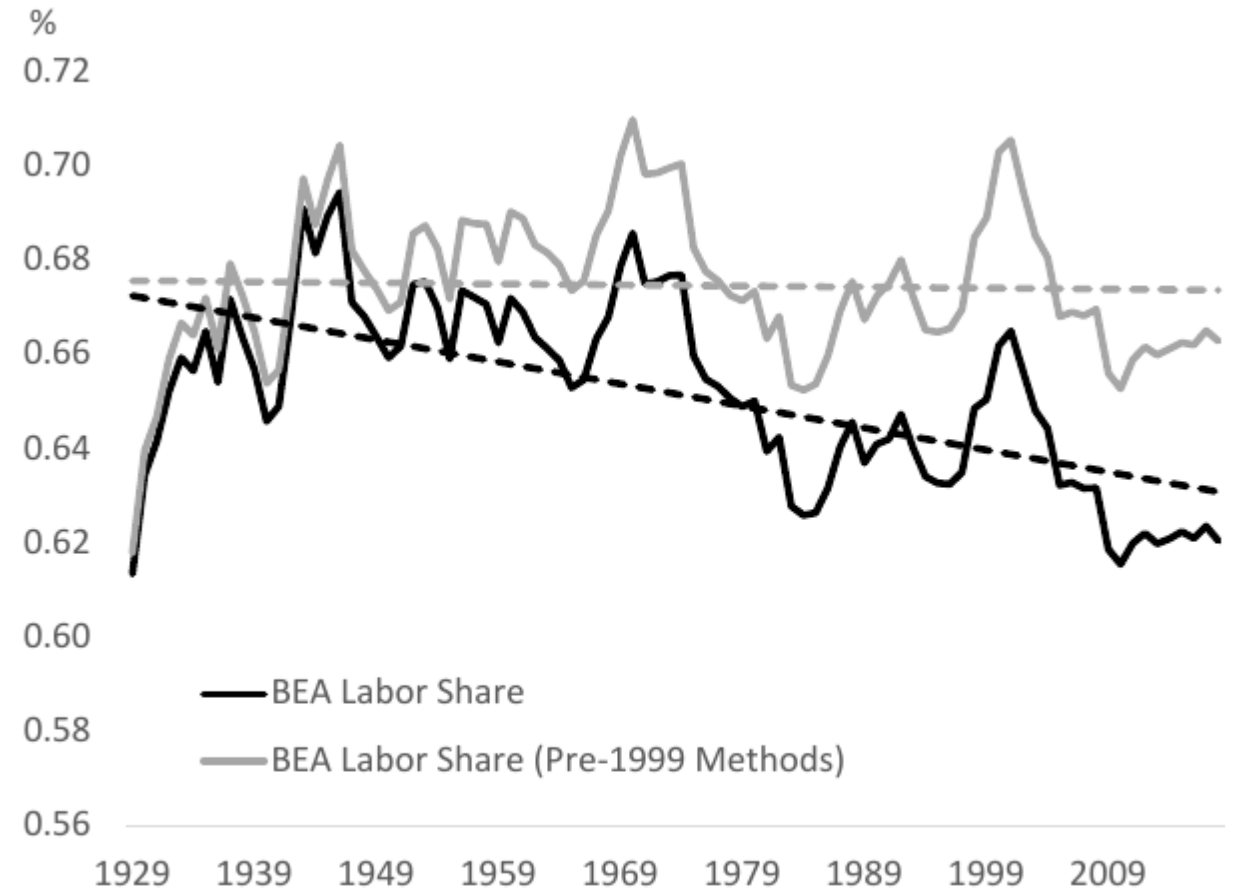


Figure 5: Economy-Wide U.S. Labor Share

# Is Labor Share Falling?

- We are interested in welfare of workers and owners of capital
  - How much they can afford to consume
- GDP not appropriate for this:
  - Owners of capital must maintain capital (can't consume that part)
  - Must also build new capital (can't consume that part)
- Net domestic product (NDP) more appropriate for labor share calculations (takes out depreciation)
- Role of high depreciation capital has been increasing (e.g., software) which helps explain downward trend for labor share of GDP

# Better and Cheaper Machines?

- One reason why labor share may be falling: better or cheaper machines
- Doesn't work with Cobb-Douglas production function

$$Y = A(zK)^a L^{1-a}$$

- In this case:

$$r = az^a AK^{a-1} L^{1-a} = a \frac{Y}{K}$$

$$rK = aY$$

- Labor share independent of  $z$

# More General Production Function

- Turns out to be very knife-edge
- Constant Elasticity of Substitution (CES) production function:

$$Y_t = \left[ a(A_{K,t}K_t)^{\frac{\sigma-1}{\sigma}} + (1-a)(A_{L,t}L_t)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Generalization of Cobb-Douglas
- Key parameter:
  - $\sigma$  is elasticity of substitution between capital and labor

# Elasticity of Substitution

$$Y_t = \left[ a(A_{K,t}K_t)^{\frac{\sigma-1}{\sigma}} + (1-a)(A_{L,t}L_t)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- $\sigma$  is elasticity of substitution between capital and labor
  - If price of capital falls by 1% relative to price of labor, quantity of capital used in production increases by  $\sigma\%$  relative to quantity of labor (slope of relative factor demand curve)
  - $\sigma = 1$  for Cobb-Douglas
  - $\sigma > 1$ : Labor and capital “gross substitutes”
  - $\sigma < 1$ : Labor and capital “gross complements”

# Labor Share in CES Case

$$s_{L,t} = 1 - a^{\sigma} \left( \frac{A_{K,t}}{r_t} \right)^{\sigma-1}$$

- $\sigma$  key parameter. If  $\sigma = 1$  back in Cobb-Douglas case
- $\sigma > 1$ :
  - Increase in  $A_{K,t}$  and decrease in  $r_t$  reduce the labor share (increase the capital share)
  - In this case, improvements in capital imply that firms use a lot more capital, increasing the capital share

# Better and Cheaper Machines

- Machines have been getting cheaper( $r_t$  has been falling)
- Perhaps this is why labor share has been falling?
- Only works if  $\sigma > 1$
- But whether this is the case is not clear.

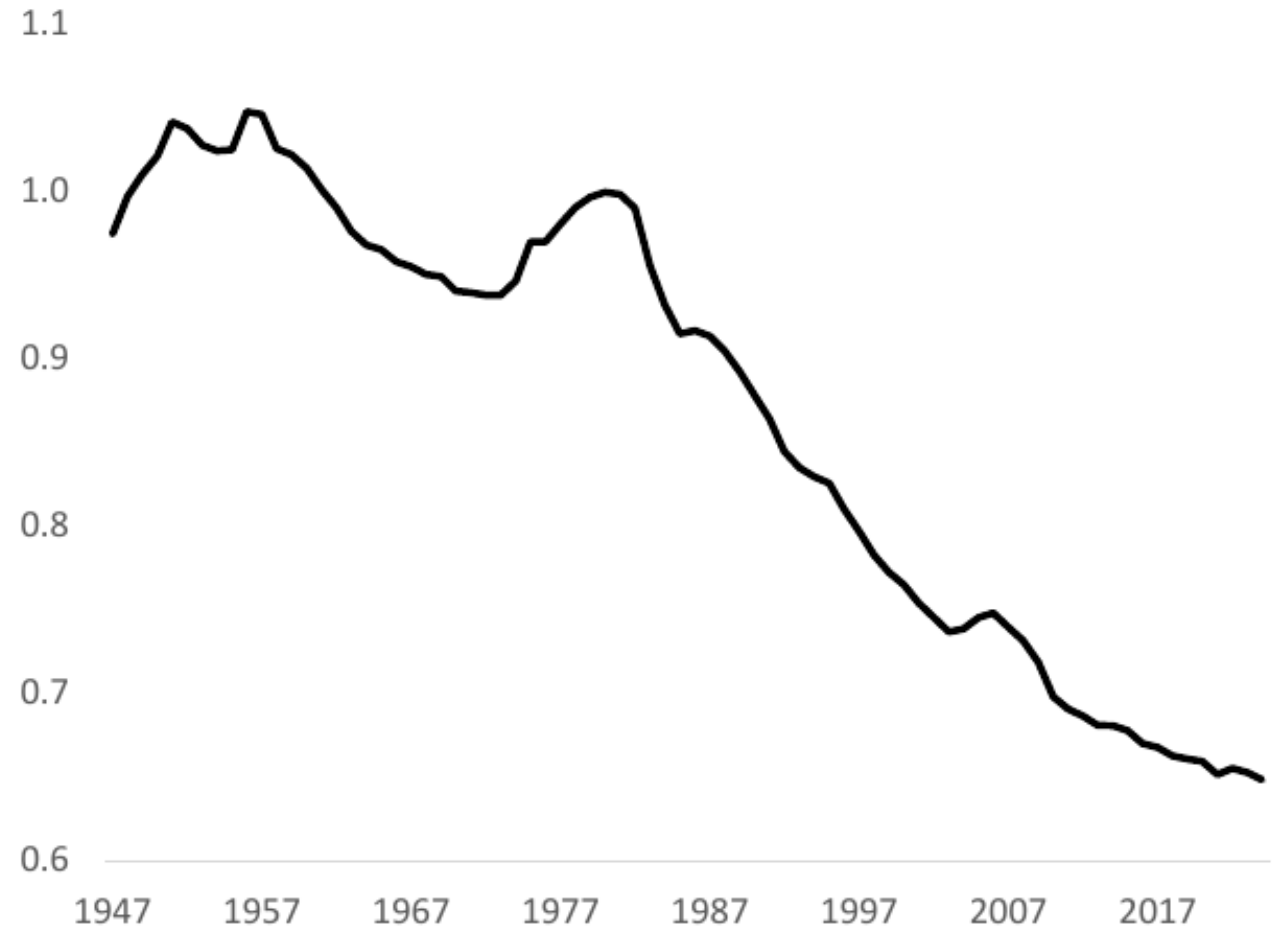
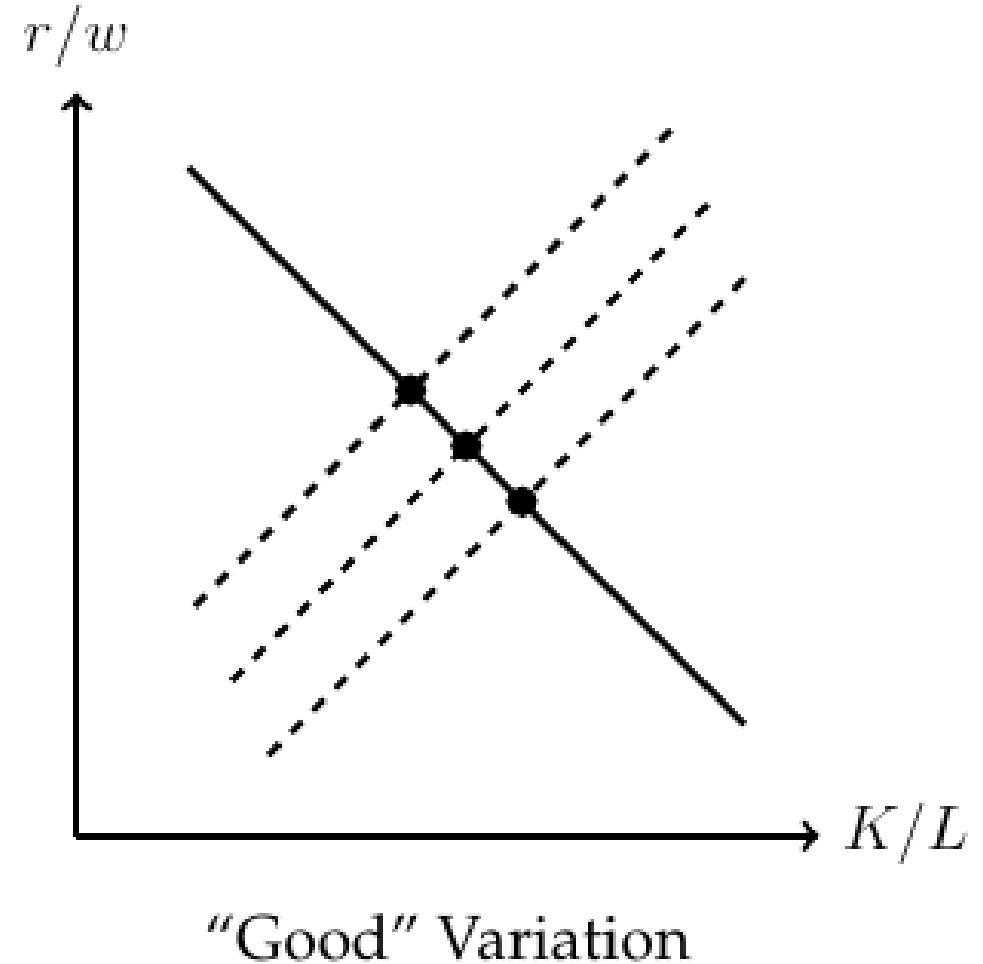


Figure 6: The Price of Investment Goods Relative to the Price of Output

# Estimating $\sigma$ is Hard

Two problems:

- Simultaneous equations problem: Need pure variation in relative supply curve to estimate slope of relative demand curve
- Aggregate  $\sigma$  will be larger than firm-level  $\sigma$  (Houthakker, 1955)
- Most estimate of  $\sigma$  suggest it is smaller than 1



# Wages and Technology with CES

- Wage with CES production:

$$w = (1 - a)A_L^{\frac{\sigma-1}{\sigma}} \left(\frac{Y}{L}\right)^{\frac{1}{\sigma}}$$

- Change in  $A_K$  and  $K$ :

$$\frac{\partial w}{\partial A_K} = \frac{a(1 - a)}{\sigma} A_L^{\frac{\sigma-1}{\sigma}} A_K^{-\frac{1}{\sigma}} Y^{-1} L^{\frac{-1}{\sigma}} K^{\frac{\sigma-1}{\sigma}} > 0$$

$$\frac{\partial w}{\partial K} = \frac{a(1 - a)}{\sigma} A_L^{\frac{\sigma-1}{\sigma}} A_K^{\frac{\sigma-1}{\sigma}} Y^{-1} L^{\frac{-1}{\sigma}} K^{\frac{-1}{\sigma}} > 0$$

# Technology Displacing Workers

- Both Cobb-Douglas and CES imply that technology and capital make workers better off
- Don't make contact with notion that technology may **displace** workers
- We need a different type of production function for this

# Task-Based Production Function

- Production involves completion of a range of tasks
- We now model production in this way
- Suppose production involves completing  $J$  tasks:

$$Y = \left[ \sum_{j=1}^J y(j)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- $y(j)$  denote number of time task  $j$  is completed
- Tasks are combined using a CES production function

# Production of Tasks

- Tasks are performed either with labor or machines:

$$y(j) = \psi_L(j)L(j) + \psi_K(j)K(j)$$

- $L(j)$  and  $K(j)$ : amount of labor and capital used to produce task  $j$
  - $\psi_L(j)$  and  $\psi_K(j)$ : productivity of labor and capital in producing task  $j$
- Labor and capital are perfect substitutes in performing task  $j$

# How Is Task $j$ Performed?

- Cost of producing task  $j$  with labor:  $\frac{w}{\psi_L(j)}$
- Cost of producing task  $j$  with capital:  $\frac{r}{\psi_K(j)}$
- Firm will do whichever is cheaper
- Will produce task  $j$  with labor if

$$\frac{w}{\psi_L(j)} < \frac{r}{\psi_K(j)}$$

and otherwise with capital

# Task-Based Production: An Example

- Good produced with 7 tasks
- Ordered by comparative advantage of labor
- Tasks 1, 2, 3 performed by capital
- Tasks 4, 5, 6, 7, performed by labor

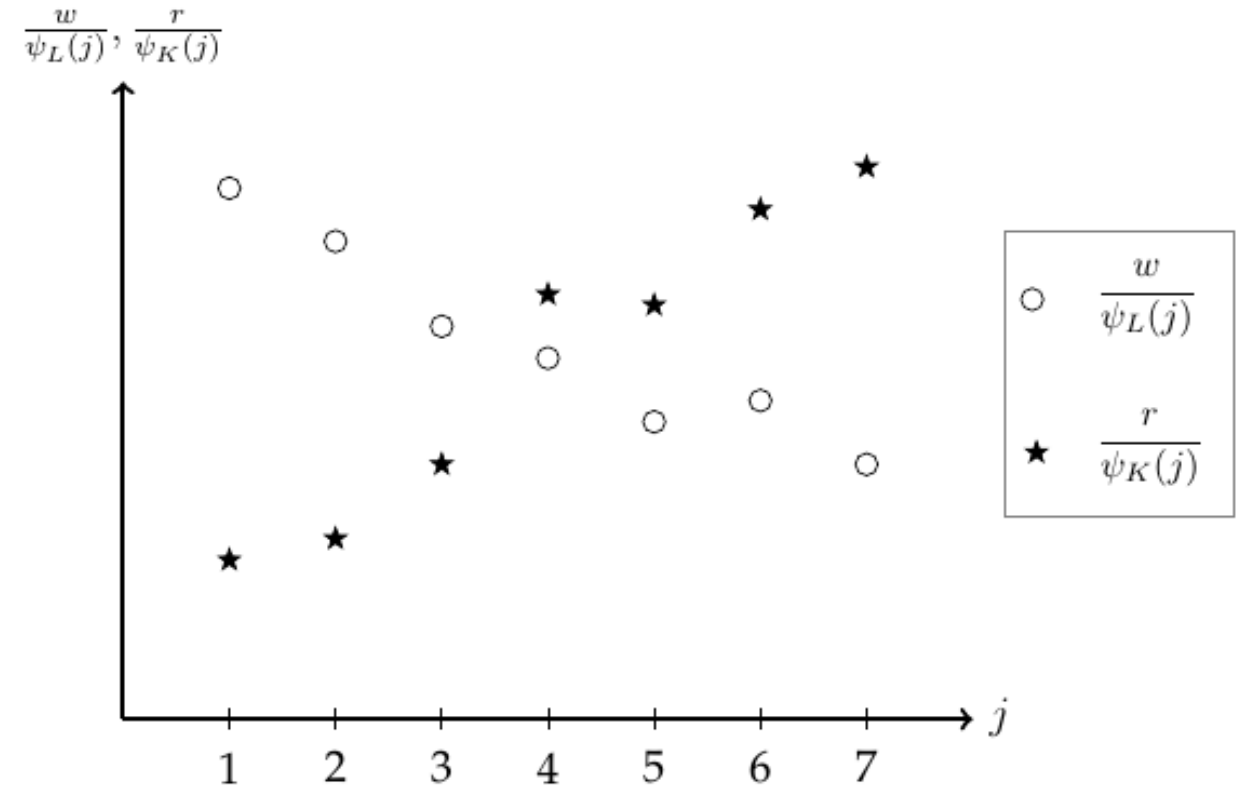


Figure 8: Cost of Producing Tasks with Labor and Capital

# Three Kinds of Technical Progress

- Innovation A:
  - Cost of performing task 2 falls
  - Machines perform task both before and after innovation
  - No workers lose their jobs
  - Cost of producing good falls
  - Raises real wages of workers (purchasing power of wages)
  - Workers better off
  - Pure **productivity effect**

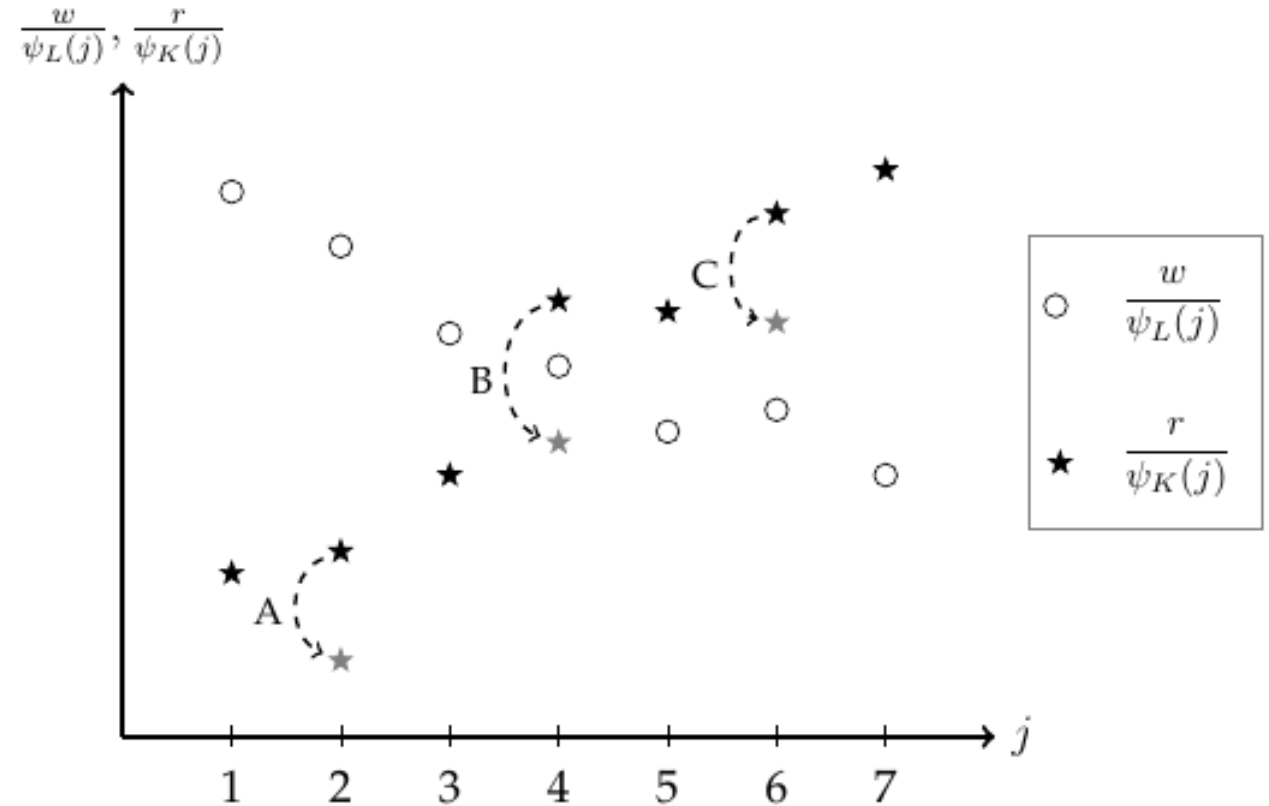


Figure 9: Three Types of Technological Change

# Three Kinds of Technical Progress

- Innovation C:
  - Machine better at performing task 6
  - But still worse than labor
  - Labor performs task both before and after innovation
  - Innovation has no immediate effect on the economy

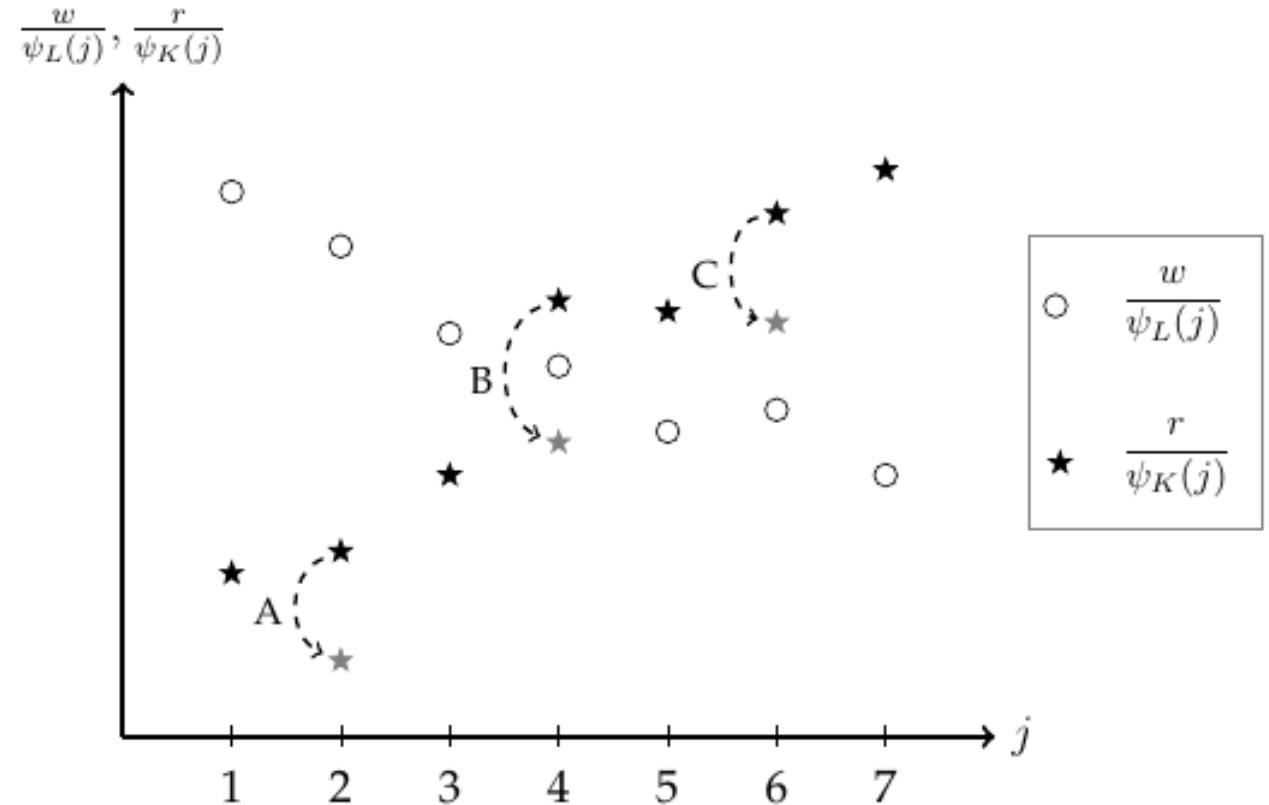


Figure 9: Three Types of Technological Change

# Three Kinds of Technical Progress

- Innovation B:
  - Machine becomes better at performing task 4
  - Firm switches from labor to capital for task 4
  - Workers lose their jobs
  - This **displacement effect** hurts workers
  - Also has a productivity effect like innovation A

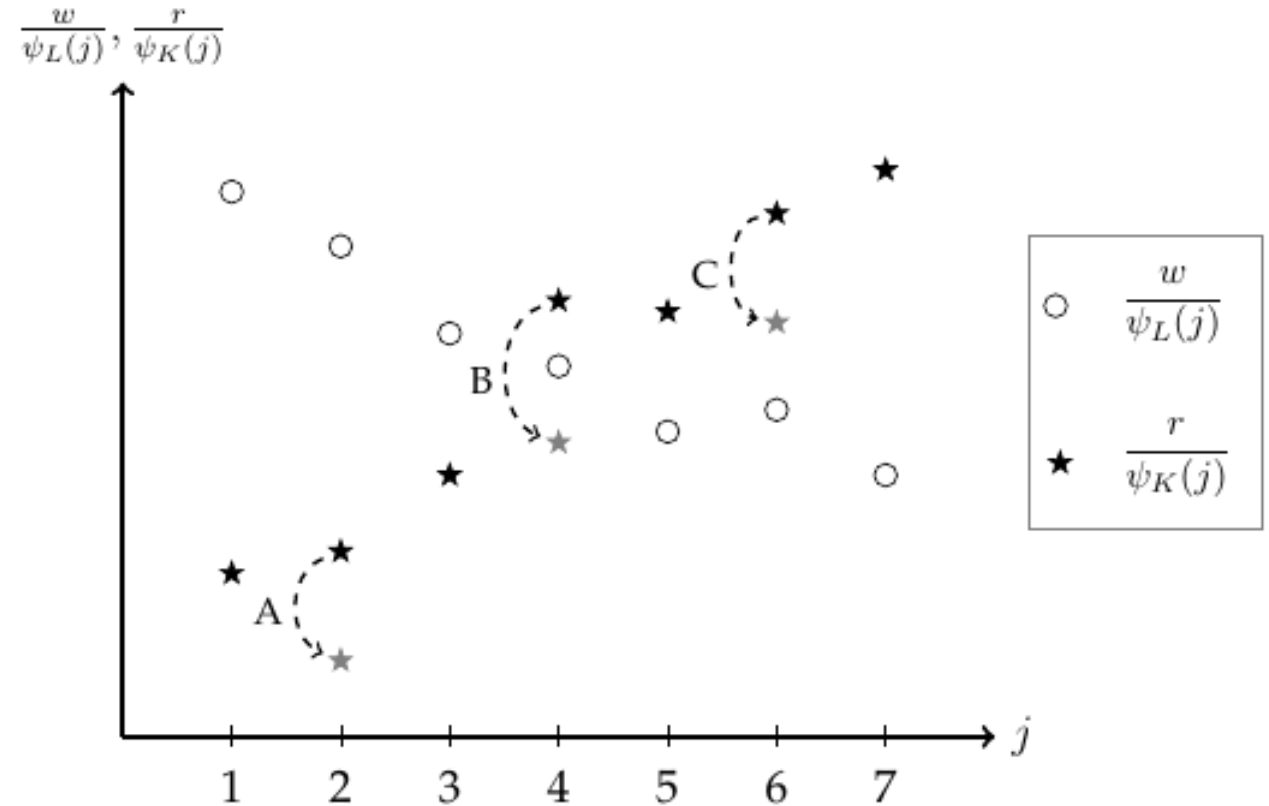


Figure 9: Three Types of Technological Change

# Displacement Effect Lowers Wages

- Automation of task 4 reduces labor demand and increases capital demand
- Real wages fall
- Rental rate on capital increases

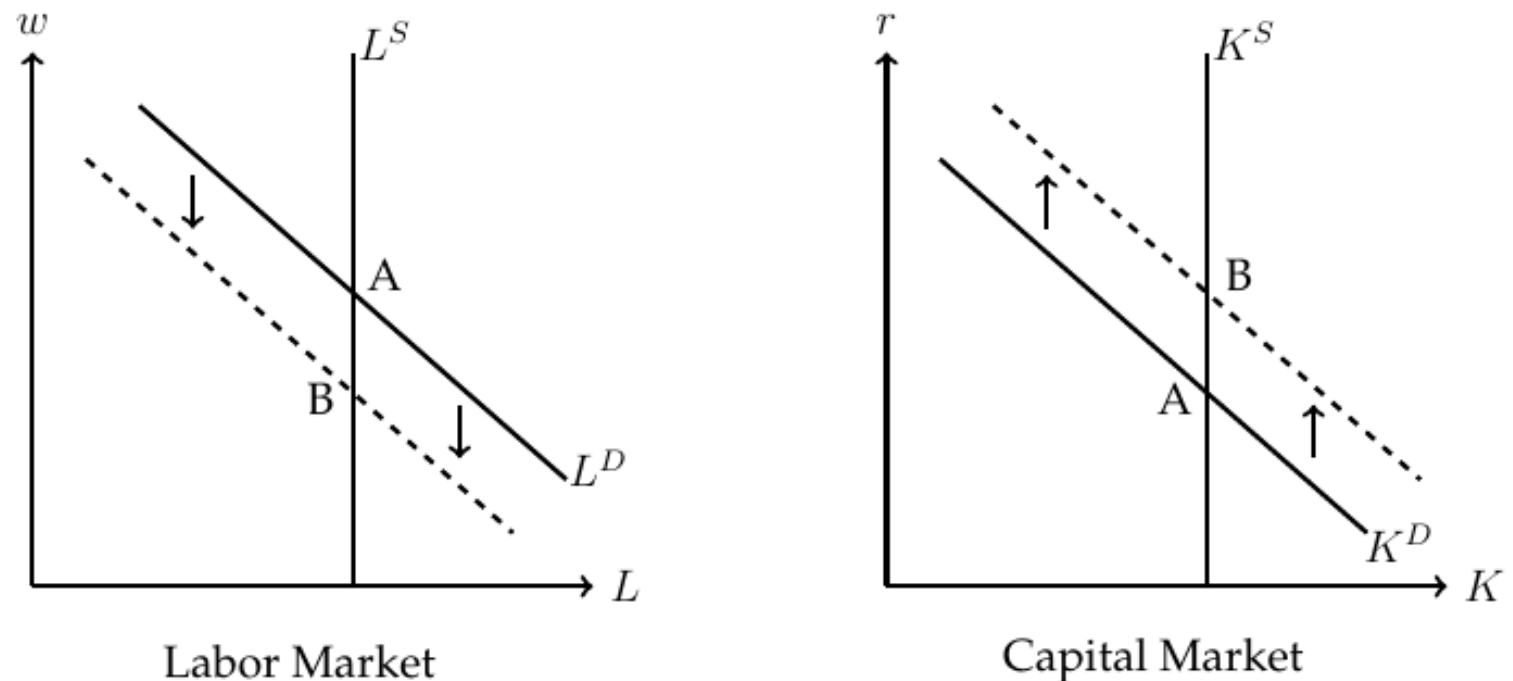


Figure 10: Effect of Automation on Factor Demand

# Aggregate Effect of Large Innovations

- Innovation can cause enormous cost reductions:
  - 18<sup>th</sup> century: Cost of spinning cotton fell by more than two orders of magnitude
  - 20<sup>th</sup> century: Cost of computing fell by more than six orders of magnitude
  - 21<sup>st</sup> century: AI/robotics will dramatically reduce costs for various tasks
- How do such huge innovations affect aggregate output?

# Aggregate Effect of Large Innovations

- Depends critically on how substitutable tasks are in production:

$$Y_t = \left[ a(A_{K,t}K_t)^{\frac{\sigma-1}{\sigma}} + (1-a)(A_{L,t}L_t)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Think of  $A_{K,t}K_t$  as bundle of tasks performed by capital
- Suppose  $A_{K,t} \rightarrow \infty$  (but  $A_{L,t}$  unchanged).
- How does this affect  $Y_t$ ?

# Weak Links

- Suppose  $\sigma = 1/2$ .
- Then  $\frac{\sigma-1}{\sigma} = -1$  and production function becomes

$$\frac{1}{Y_t} = \frac{a}{A_{K,t}K_t} + \frac{1-a}{A_{L,t}L_t}$$

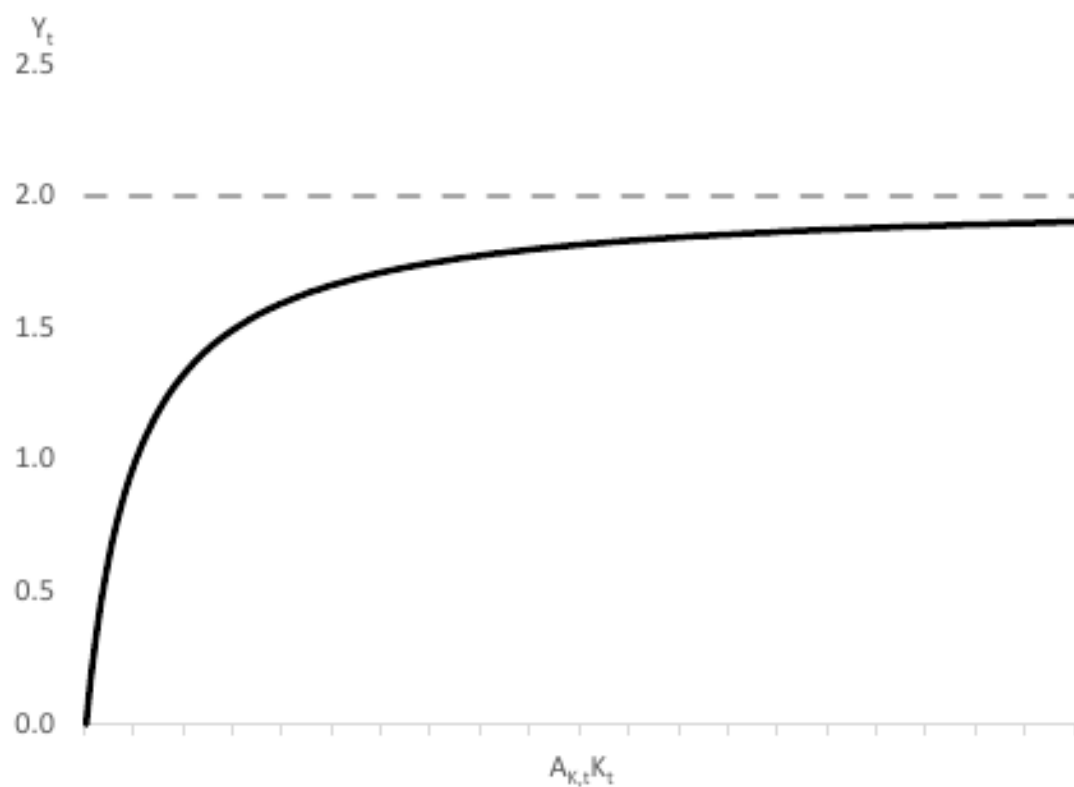
- If  $A_{K,t}K_t \rightarrow \infty$  then  $\frac{a}{A_{K,t}K_t} \rightarrow 0$ .

So:

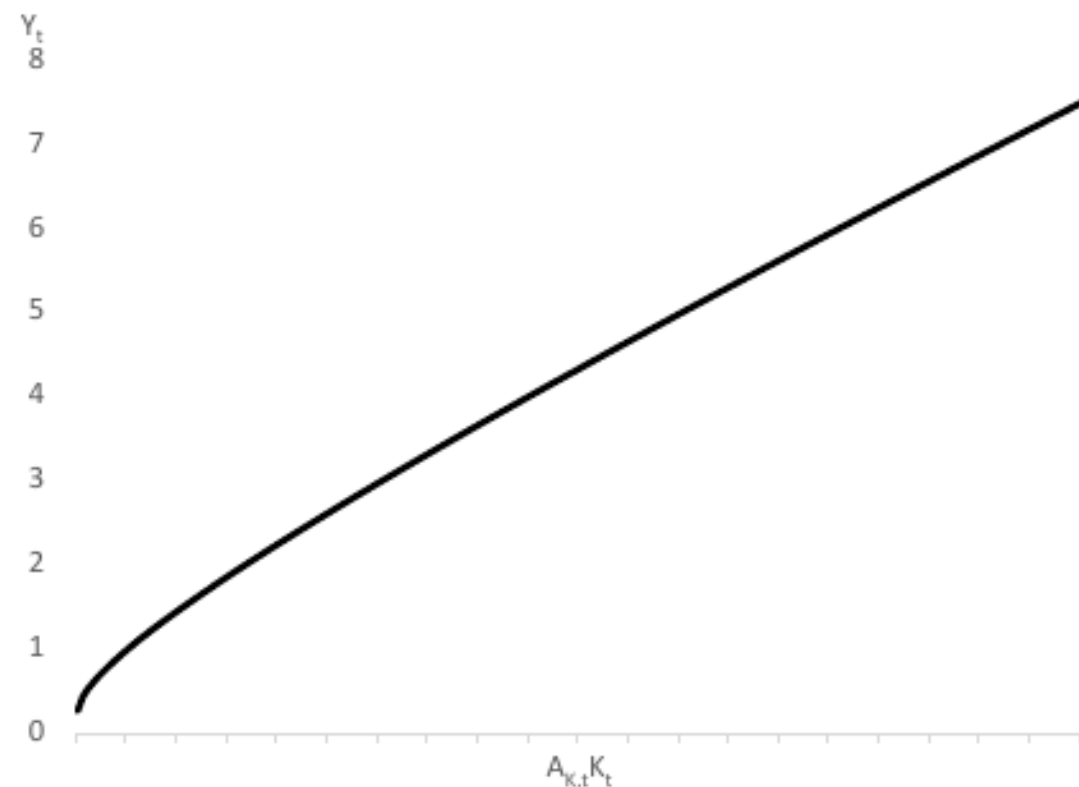
$$Y_t = \frac{A_{L,t}L_t}{1-a}$$

- Capital becomes infinitely productive. But output remains finite.
- Labor is a “weak link” (bottleneck)
- Since it is hard to substitute capital for labor, economy runs into rapidly diminishing returns to capital

# Weak Links



(a)  $\sigma = 0.5$



(b)  $\sigma = 2$

Figure 11: Weak Links and Aggregate Output

# Aggregate Effect of Large Innovations

- How much  $Y_t$  increases depends on initial cost share of tasks exposed to innovation  $s_{K,t}$
- Let  $\tilde{Y}_t$  denote output when  $A_{K,t}K_t \rightarrow \infty$
- Increase in output from infinitely productive capital:

$$\frac{\tilde{Y}_t}{Y_t} = \left( \frac{1}{1 - s_{K,t}} \right)^{\frac{\sigma}{1-\sigma}}$$

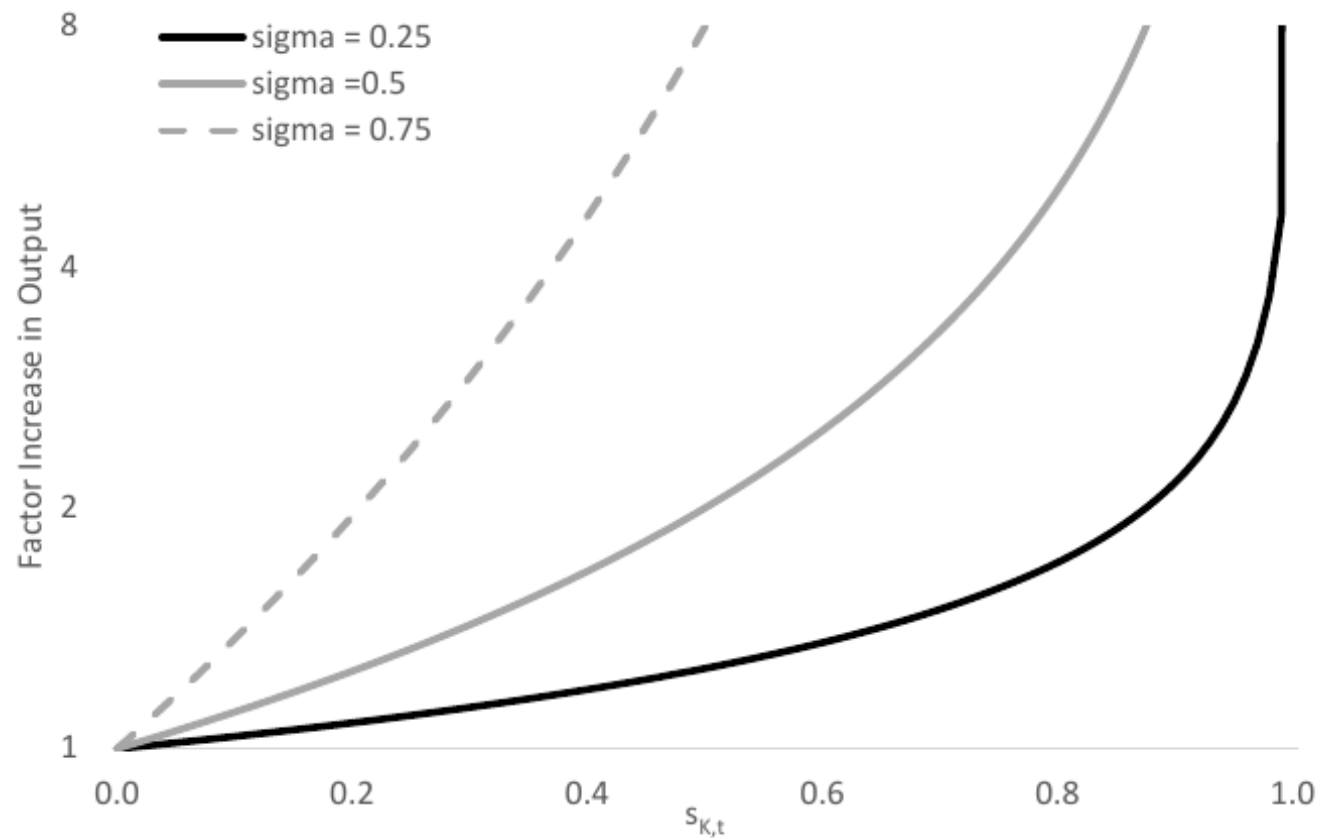


Figure 12: Increase in Output if Capital Becomes Infinitely Productive

# Will Technology Make Labor Irrelevant

Two forces:

1. Automation of tasks:

- This tends to lower the labor share of income

2. Continued improvement of machines at what they already do

- If  $\sigma < 1$ , this tends to raise the labor share
- Task prices fall faster than quantity rises
- This is an example of Baumol's disease

- Not clear that first force will be stronger than second